3.7.5 Half-Angle formula

Theorem 3.6: In $\triangle ABC$

(i) $\sin\dfrac{A}{2}=\sqrt{\dfrac{(s-b)(s-c)}{bc}}$, (ii) $\cos\dfrac{A}{2}=\sqrt{\dfrac{s(s-a)}{bc}}$,

(iii) $\tan\dfrac{A}{2}=\sqrt{\dfrac{(s-b)(s-c)}{s(s-a)}}$, where s is the semi-perimeter of $\triangle ABC$ given by $s=\dfrac{a+b+c}{2}$

Proof. (i)

$$\begin{aligned}\sin\frac{A}{2} &= +\sqrt{\sin^2\frac{A}{2}}=\sqrt{\frac{1-\cos A}{2}}=\sqrt{\frac{1}{2}\left(1-\frac{b^2+c^2-a^2}{2bc}\right)}\\ &= \sqrt{\frac{2bc-b^2-c^2+a^2}{4bc}}=\sqrt{\frac{a^2-(b-c)^2}{4bc}}\\ &= \sqrt{\frac{(a+b-c)(a-b+c)}{4bc}}=\sqrt{\frac{(a+b+c-2c)(a+b+c-2b)}{4bc}}\\ &= \sqrt{\frac{(2s-2b)(2s-2c)}{4bc}}=\sqrt{\frac{(s-b)(s-c)}{bc}}\end{aligned}$$

Thus, $\sin\dfrac{A}{2}=\sqrt{\dfrac{(s-b)(s-c)}{bc}}$.

Similarly, we can prove the other two results.

The other half-angle formulas are

$$\begin{aligned}\sin\frac{B}{2} &= \sqrt{\frac{(s-c)(s-a)}{ac}}, & \sin\frac{C}{2} &= \sqrt{\frac{(s-a)(s-b)}{ab}}\\ \cos\frac{B}{2} &= \sqrt{\frac{s(s-b)}{ac}}, & \cos\frac{C}{2} &= \sqrt{\frac{s(s-c)}{ab}}\\ \tan\frac{B}{2} &= \sqrt{\frac{(s-a)(s-c)}{s(s-b)}}, & \tan\frac{C}{2} &= \sqrt{\frac{(s-a)(s-b)}{s(s-c)}},\end{aligned}$$

Corollary: $\sin A=2\sin\dfrac{A}{2}\cos\dfrac{A}{2}=2\sqrt{\dfrac{(s-b)(s-c)}{bc}}\sqrt{\dfrac{s(s-a)}{bc}}$

$$\sin A=\frac{2}{bc}\sqrt{s(s-a)(s-b)(s-c)}$$

Area of a triangle (Heron's Formula)

Heron's formula is named after Hero of Alexendria, a Greek Engineer and Mathematician in (CE 10 - 70). This formula is used only when all the three sides are known.

Theorem 3.7: In $\triangle ABC$, $\triangle=\sqrt{s(s-a)(s-b)(s-c)}$ where s is the semi-perimeter of $\triangle ABC$.

Proof.

$$\begin{aligned}\triangle &= \frac{1}{2}ab\sin C=\frac{1}{2}ab\left(2\sin\frac{C}{2}\cos\frac{C}{2}\right)\\ &= ab\sqrt{\frac{(s-a)(s-b)}{ab}}\sqrt{\frac{s(s-c)}{ab}}=\sqrt{s(s-a)(s-b)(s-c)}\end{aligned}$$